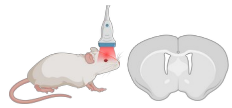

deConfUSion fUSI Studio

Full Updated User Manual

Data analysis pipeline for functional ultrasound imaging

Author / pipeline owner	Soner Caner Cagun
Manual refresh	Generated from uploaded source package on 09 June 2026
Source package inspected	HUMOR-Analysis-Tool (2)(9).zip
Current active source summary	80 root MATLAB files + 2 atlas_tools MATLAB files, excluding backups and .git.
Naming note	User-facing manual name: deConfUSion. Current MATLAB helper prefix in the ZIP: deConfUSIon_*.
Scope note: This manual is based on static inspection of the uploaded source package. It documents how the toolbox is intended to be used and what the current files implement, but it is not a substitute for running MATLAB validation on representative 2D, step-motor, and 3D/matrix datasets before publication.	
Renaming note: The toolbox has been renamed in this manual from HUMoR to deConfUSion. The repository folder, old PDF, old README passages, comments, and some archived/backup material may still contain HUMoR/HUMOR names. Do not interpret those legacy names as separate active toolboxes.	

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1. What this manual teaches

This manual explains how to analyze functional ultrasound imaging data with deConfUSion / fUSI Studio. It is written for a new user who has a raw or preprocessed fUSI dataset and wants to move from data loading to quality control, preprocessing, percentage signal change analysis, signal change map visualization, masking, atlas registration, segmentation, functional connectivity, and group-level outputs.

By the end of the manual, the user should be able to launch the current split-GUI version safely, load 2D, step-motor/multi-slice, or matrix/3D-style data, confirm TR and dataset timing, run QC, apply only justified corrections, create PSC/SCM outputs, save masks, inspect atlas registration, extract region time courses, and prepare group-level summaries.

> **Tip / Trick:** Do not treat the GUI as a black box. Every preprocessing step changes the dataset. Keep the raw dataset untouched, use the active dataset dropdown carefully, and export the Studio Log at the end of every analysis session.

2. Current toolbox name and source package status

The current user-facing toolbox name is deConfUSion. In the uploaded ZIP, the main convenience launcher is `deConfUSIon.m`, and many helper files use the `deConfUSIon_` prefix. The repository folder and some historical material still use HUMoR/HUMOR names because the project was originally named HUMoR fUSI Studio.

Current active launcher architecture:

```
cd('D:\Github\HUMOR-Analysis-Tool')
run_fusi_studio
% or
deConfUSIon
```

- `deConfUSIon.m` is a short convenience wrapper that adds the toolbox root and `atlas_tools` to the MATLAB path, then calls `run_fusi_studio`.
- `run_fusi_studio.m` is the recommended launcher. It assembles `fusi_studio_GUI.m` and `fusi_studio_callback.m` into a temporary runtime file outside the toolbox root.
- The current ZIP does not contain a root-level `fusi_studio.m`. Therefore, old instructions that say to run `fusi_studio` directly are outdated for this package.
- The current active GUI title is `deConfUSIon` and contains 10 visible sections. Sections 8-10 are currently placeholder/reserved analysis areas.

> **Current-source note:** The README inside the uploaded package still contains some older references, such as `fusi_studio.m`, `studio_load_options_dark_dialog_patch16.m`, and several registration helper filenames that are no longer present at root level. This manual follows the current active files in the ZIP, not the older README wording.

Current development status

Workflow area	Status in current ZIP
2D probe / single-slice workflow	Most complete and primary active workflow.
Step-motor / 2D multi-slice workflow	Available; motor reconstruction, segmentation, and FC helper support are present, but downstream group analysis and FC should still be validated carefully.
3D / matrix probe workflow	Partially supported by loader, PSC, registration, segmentation, and some visualization paths; advanced analysis should be validated per dataset.
Functional connectivity	Available as a large standalone GUI with seed/ROI/pair/graph analysis, step-motor helpers, JM ordering support, and group FC backends.
Group analysis	Available for ROI/map workflows, with FC backends and Fisher-z group statistics patches present in current source.
StimBox / PulsePal / motor acquisition scripts	Included as optional hardware-dependent scripts; they require local configuration and are not necessary for offline analysis.

3. Abbreviations and key terms

Abbreviation / term	Meaning in this toolbox
deConfUSion	Current user-facing toolbox name in this manual. Current code prefix appears as `deConfUSIon_`.
HUMoR / HUMOR	Legacy project/toolbox name. It may still appear in paths, archived files, README text, or old manuals.
fUSI	Functional ultrasound imaging.
PD / Power Doppler	Power Doppler-like signal intensity used as the fUSI signal source.
CBV	Cerebral blood volume. fUSI power Doppler is often interpreted as CBV-like hemodynamic signal.
TR	Repetition time between frames/volumes, in seconds.
PSC	Percentage signal change: 100 x (signal - baseline) / baseline.
SCM	Signal change map; spatial map of PSC or related signal change.
ROI	Region of interest.
QC	Quality control; diagnostic analysis before and after processing.
tSNR	Temporal signal-to-noise ratio.
CNR	Contrast-to-noise ratio.
DVARS	Frame-to-frame signal-change metric used for motion/artifact detection.
COM	Center of mass; used for apparent motion/drift QC.
PCA	Principal component analysis; used for QC and optional denoising.
ICA	Independent component analysis; used for optional denoising.
FC	Functional connectivity; correlation-based relationship between seed/ROI time courses.
Fisher z	atanh(r) transform used when averaging/testing correlation values.
GLM	General linear model. Current GUI has placeholder sections for future GLM/regression workflows.
Allen atlas / CCF	Allen Common Coordinate Framework / atlas resources used for registration and anatomical labels.
Reg2D	Saved 2D coronal registration object used for 2D/step-motor atlas workflows.
JM order/color	Atlas color/order support using `atlas_tools/rgb2acr.xlsx` and `list_selected_regions.txt`.
StimBox / PulsePal	Optional external stimulation/trigger hardware interfaces.
Underlay	Anatomical/background image beneath overlays in SCM/video/FC viewers.
Overlay / signal mask	Mask limiting displayed/analyzed signal region.
Brain / underlay mask	Mask limiting anatomical underlay or valid brain area.

4. Core concepts and calculations

Concept	Meaning / caution
fUSI signal	Power Doppler-like intensity changes. Interpret as hemodynamic/CBV-like, not direct neuronal activity.
Active dataset	Dataset currently selected in the dropdown. Most mistakes happen when the active dataset is not the intended version.
Baseline window	Time interval used to compute baseline intensity. It must be stable and before stimulus/injection or major motion.
Processing chain name	Readable dataset label that should preserve animal/scan identity and processing steps. The current package includes helper files to preserve these names after reloads.
Step-motor reconstruction	Frame-based conversion of continuous or split slice acquisitions into [Y X Z T] data.
Atlas-space segmentation	Extraction of region-wise time courses using registered atlas label maps; not AI segmentation.

4.1 PSC calculation

For a voxel or pixel with intensity I(t), the baseline B is computed from selected baseline frames. PSC is then:

```
PSC(t) = ((I(t) - B) / B) x 100
```

The same logic applies to 2D time series [Y X T] and multi-slice or matrix data [Y X Z T]. ROI time courses are generated by averaging PSC values inside the selected ROI or anatomical label at each time point.

Tip / Trick: A bad baseline contaminates every PSC frame. Before exporting maps, inspect the raw movie, global signal, and

ROI trace around the baseline window.

5. Installation and setup

1. Install MATLAB. Current source comments repeatedly target MATLAB 2017b and MATLAB 2023b compatibility.
2. Install required MATLAB toolboxes where available. Image Processing Toolbox is especially useful for registration, masks, filtering, resizing, and morphology. Some code has fallbacks, but not every workflow can be fully toolbox-free.
3. Unzip or clone the toolbox into a local working folder. Windows is currently the preferred platform.
4. Keep old backup folders out of the active MATLAB path. In this ZIP, the `backups` folder contains historical patches and compressed cleanup packages; do not add it recursively to the path unless debugging intentionally.
5. Make sure `allen_brain_atlas.mat` is present in the toolbox root if atlas registration/segmentation is needed.
6. Start MATLAB, change to the toolbox root, and run `run_fusi_studio` or `deConfUSIon`.

```
cd('D:\Github\HUMOR-Analysis-Tool')
run_fusi_studio
```

Path warning: Do not run old patch scripts from backups during normal use. They are kept for recovery/history, not for active analysis.

6. Expected data and folder organization

Input type	Expected structure	Notes
.mat	[Y X T] or [Y X Z T]	Best-supported route. Preferred variable is `I`, but loader searches common variables and nested candidates. Metadata such as TR, time vectors, masks, and acquisition parameters can be passed through.
.nii	2D/3D/4D NIfTI	Supported by `niftiread`. Always verify orientation, dimensions, TR, and total duration manually.
.nii.gz	Compressed NIfTI	Gunzip/read workflow. Same caution as .nii.
Step-motor split files	Many MAT files grouped by slice/time	Use Motor reconstruction. Consistent names such as `slice1_t001.mat`, `slice2_t001.mat` help automatic grouping.
Registered/segmentation MATs	Functional data + registration/label information	Used by atlas/segmentation and FC workflows; verify matching slice/space before trusting labels.

6.1 TR and probe defaults

The current loader attempts to infer TR from explicit seconds fields, millisecond fields, sampling rates, or timestamp vectors. It initializes `TRWasImputed` and stores the detection source in metadata. If no usable TR exists, probe-type defaults are used and the GUI asks the user to confirm.

Probe/data type inferred by loader	Default TR if not found
2D Probe	0.320 s
Matrix (3D) Probe	0.480 s

6.2 Recommended folder structure

```
RawData/
  <project>/<animal>/<session>/<scan>.mat

AnalysedData/
  <project>/<animal>/<session>/<scan>/
    QC/
    Preprocessing/
    PSC/
    Visualization/
    Masks/
    Registration/
    Registration2D/
    Segmentation/
```

```
FunctionalConnectivity/
GroupAnalysis/
Reports/
```

Tip / Trick: Never overwrite raw data. Every preprocessing step should create a new dataset version. Always check the dropdown before SCM, Video, Registration, Segmentation, FC, or Group Analysis.

7. Step-by-step individual analysis workflow

Step	Action
0. Prepare analysis session	Open MATLAB in the toolbox folder, confirm the acquisition log, TR, stimulus/injection timing, acquisition mode, and expected baseline window. Keep raw data read-only.
1. Launch	Run `run_fusi_studio` or `deConfUSIon`. The split GUI is assembled into a temporary runtime and the main deConfUSion window opens.
2. Load fUSI Data	Select .mat, .nii, or .nii.gz. Confirm probe type and TR. Check active dataset label.
3. Run QC before preprocessing	Use Full QC for normal datasets or Specific QC for selected modules. Review plots before data modification.
4. Apply recommended processing only when justified	Use frame rejection/scrubbing for bad frames, imregdemons for drift/spatial instability, and Motor for step-motor/multi-slice reconstruction.
5. Apply advanced processing only when needed	Temporal smoothing, filtering, PCA/ICA, and despiking can improve readability but can remove true signal. Save/report all settings.
6. Inspect with Time-Course Viewer	View raw/PSC dynamics, raw movie, and global/ROI signal stability.
7. Create SCM maps	Select baseline, underlay, polarity, caxis, alpha modulation, smoothing, slice, and masks. Export maps and ROI time courses only after visual checks.
8. Validate with Video & SCM Mask	Confirm that SCM findings evolve plausibly over time and are not driven by one bad frame.
9. Draw/load masks	Use Mask Editor to draw brain/underlay and overlay/signal masks. Save bundles with clear dataset-specific names.
10. Register and segment	Register to atlas after clean anatomy/underlay exists. Inspect registration visually. Then run atlas-based segmentation where appropriate.
11. Run advanced analyses	Run FC or Group Analysis only after individual datasets, masks, labels, and exports have passed QC.
12. Export and log	Export the Studio Log, ROI text files, figures, SCM/FC outputs, segmentation tables, and any group summaries.

8. Detailed GUI and module guide

8.1 Main GUI sections

GUI section	Buttons	Purpose / status
1. Dataset	Load fUSI Data	Start here; loads .mat/.nii/.nii.gz and controls active dataset dropdown.
2. QC & Data Overview	Full QC; Specific QC	Diagnose data quality before modifying it.
3. Recommended Processing	Frame Rejection; Imregdemons; Scrubbing; Motor	Use when QC or acquisition mode justifies correction/reconstruction.
4. Advanced Processing	Temporal Smoothing/Subsampling; Filtering; PCA / ICA; Despike	Use only with documented rationale; may alter biology.
5. Visualization	Time-Course Viewer; SCM; Video & SCM Mask; Mask Editor	Inspect raw/PSC dynamics, create maps, draw/load masks, export figures.
6. Coregistration	Registration to Atlas; Segmentation	Register data/underlays to atlas and extract region-wise time courses.
7. Advanced Analysis	Functional connectivity; Group analysis	Use only after individual datasets pass QC and masks/labels are checked.
8. GLM / Regression	General Linear Models; Regression	GUI placeholders in current source. Treat as not validated until callbacks are wired.
9. Velocity Analysis	Velocity Maps; Flow / Velocity QC	GUI placeholders in current source. Not a primary validated workflow yet.
10. Community Analysis	Placeholder Community A; Placeholder Community B	Reserved/placeholder section.

GUI status: The current GUI source creates 10 sections. Sections 1-7 are the main analysis workflow. Sections 8-10 are visible but are placeholders/reserved in the current source and should not be treated as validated analysis modules.

8.2 Detailed module guide

Button/module	Main code	What it does / caution
Load fUSI Data	loadDataCallback.m, loadFUSIData.m, studio_load_options_dark_dialog.m	Loads MAT/NIfTI, detects variables, infers TR/probe, asks for confirmation. Always verify dimensions and total time.
Dataset dropdown	datasetDropdownCallback.m, deConfUSIon_*display/name helpers	Controls active raw/processed dataset and preserves readable processing-chain names. Check after every processing step.
Full / Specific QC	qc_fusi.m, frameRateQC.m	Frequency, spatial, temporal, motion/stability, burst, tSNR, CNR, common-mode, reliability, PCA QC.
Frame Rejection	frameRateQC.m, interpolateRejectedVolumes.m	Detects abnormal frame/global-intensity events and interpolates rejected volumes.
Imregdemons	imregdemons_preprocess.m	Block mean/median plus non-rigid drift correction; supports true 3D and slice-wise step-motor mode.
Scrubbing	scrubbing.m	DVARS or global-signal bad-frame detection with interpolation and QC.
Motor	motor.m	Frame-based continuous or split step-motor reconstruction into [Y X Z T].
Temporal Smoothing/Subsampling	temporalsmoothing.m	Sliding average or block averaging/subsampling. Block mode changes effective TR.
Filtering	filtering.m	Robust Butterworth low/high/band-pass filtering for 2D+time and [Y X Z T]; restores voxelwise mean by default.
PCA / ICA	pca_denoise.m, ica_denoise.m	Interactive component inspection/removal with slice/all-slice support. Do not remove components blindly.
Despike	despike.m	Voxel-wise MAD spike detection and interpolation.
Time-Course Viewer	fUSI_Live_Studio.m	Raw/PSC movie and time-course inspection.
SCM	computePSC.m, SCM_gui.m	Computes/displays PSC maps, ROI traces, masks, underlays, exports.
Video & SCM Mask	play_fusi_video_final.m, showScmVideoSetupDialog.m	Dynamic playback with overlay/underlay/mask controls.
Mask Editor	mask.m	Draws/saves brain/underlay and overlay/signal masks.
Registration to Atlas	coreg.m, coreg_coronal_2d.m, coreg_3d.m, registration_coronal_2d.m, registration_ccf.m	Manual 2D/3D atlas registration using Allen atlas resources.
Segmentation	Segmentation.m	Atlas-based region-time extraction for registered 3D or 2D/step-motor Reg2D workflows.
Functional Connectivity	FunctionalConnectivity.m and deConfUSIon_FC_* helpers	Seed/ROI/pair/graph FC; step-motor region-name/label helpers; JM ordering support.
Group Analysis	GroupAnalysis.m, GroupAnalysis_Common.m, GroupAnalysis_Map.m, GroupAnalysis_FC.m	Group-level ROI, map, and FC analysis/export. FC statistics use Fisher-z patches in current source.

9. Quality-control workflow

Run QC before preprocessing, and rerun relevant QC after major correction steps. The current `qc_fusi.m` includes paper-style and Python-style diagnostics and saves PNG/MAT outputs where configured.

QC module	Output	Interpretation
Frequency QC	Power spectra in broad 0-2 Hz and low 0-0.1 Hz ranges.	Noise rhythms, drift, acquisition periodicity.
Spatial QC	Mean image, temporal CV, tSNR map and histogram.	Vessel visibility, spatial SNR, noisy areas.
Temporal QC	Global signal, relative global signal, DVARS, spikes.	Abrupt jumps, drift, motion-related events.
Stability QC	Intensity distribution, Gaussian/threshold style stability, rejection trace/map.	Stable acquisition support across frames/slices.
Motion QC	Center-of-mass drift.	Apparent tissue/brain movement over time.
Burst QC	Burst ratio, noisy voxel fraction, coverage over time.	Saturation-like or acquisition burst artifacts.
CNR QC	True CNR if baseline/stim windows known; otherwise pseudo CNR.	Response contrast relative to baseline variability.
Common-mode QC	Block correlation / common-mode portion.	Global artifacts dominating the signal.
Reliability QC	Finite/non-NaN support or cross-dataset reliability map.	Unreliable regions/slices and atlas-region reliability warnings.
PCA QC	Explained variance and first components, auto-saved/auto-closed.	Dominant artifact structure or abnormal components.

QC caution: QC acceptance does not mean publication-ready. It only indicates that a set of numeric checks passed. The raw movie, baseline, masks, ROI traces, and registration still require manual review.

10. Preprocessing workflow and cautions

Step	Use when	Caution
Frame rejection	When frame-rate/global-intensity QC detects abnormal frames.	Review rejected count and interpolation trace. Do not remove large parts of a dataset without documentation.
Scrubbing	Sudden motion/artifact frames detected by DVARS/global signal.	Choose detection/interpolation settings consistently across animals where possible.
Imregdemons	Slow spatial drift or deformation over time.	Median block method with nsub=100 is the recommended GUI preset. Use slice-wise mode for step-motor when appropriate.
Motor reconstruction	Continuous or split step-motor acquisitions.	Frame-based; verify frames per slice, baseline frames, trimming, and output [Y X Z T].
Temporal smoothing	Noisy time courses where smoothing is scientifically acceptable.	Sliding mode preserves length; block/subsampling mode changes effective TR.
Filtering	Known frequency band of interest or removal of drift/high-frequency noise.	Current patch supports 2D+time and [Y X Z T], returns static 2D unchanged, restores voxelwise mean, and writes higher-res QC PNGs.
PCA/ICA denoising	Identifiable artifact components.	Inspect maps and time courses. Removing biological response components invalidates interpretation.
Despiking	Sparse voxel-wise spikes.	Use sparingly and document threshold/settings.

10.1 Filtering defaults in current source

Setting	Current behavior
Type	Band-pass by default; low-pass and high-pass also supported.
FcLow	0.01 Hz default.
FcHigh	0.20 Hz default, with Nyquist safety clamping.
Order	4 by default; clamped to 1-6.
restoreMean	true by default; temporal fluctuations are filtered and voxelwise mean is restored.
chunkSize	50000 voxels default for memory-safe processing.

11. Visualization, PSC, SCM, video, and masks

Visualization is not just presentation; it is part of data validation. Use Time-Course Viewer, SCM, Video & SCM Mask, and Mask Editor together.

- Use Time-Course Viewer before SCM to inspect raw dynamics and global/ROI time courses.
- Use SCM to compute and display PSC maps. Confirm baseline, caxis, signal polarity, underlay, alpha modulation, smoothing, and slice selection.
- Use Video & SCM Mask to test whether a spatial finding is temporally plausible and not driven by one frame or a wrong underlay.
- Use Mask Editor to save brain/underlay masks and overlay/signal masks. Keep mask bundles with the dataset that generated them.
- For step-motor or multi-slice data, confirm that slice index, underlay, atlas label map, and functional slice match.

Visualization warning: SCM maps that appear strong at edges, bubbles, skull shadows, or outside valid brain masks should be treated as artifacts until proven otherwise.

12. Atlas registration and segmentation

Atlas workflows should only be performed after individual data are clean enough for anatomical interpretation. Registration is a visual step; never trust atlas labels without checking the overlay.

Workflow	Main files	Notes
2D coronal registration	\`coreg_coronal_2d.m\`, \`registration_coronal_2d.m\`	Manual registration of 2D source slice or step-motor source stack to a selected coronal atlas slice. Reg2D objects support downstream segmentation/FC.
3D registration	\`coreg_3d.m\`, \`registration_ccf.m\`, \`interpolate3D.m\`	Manual 3D atlas registration using atlas histology/vascular/regions underlays.
Atlas preparation	\`deConfUSIon_prepare_atlas.m\`, \`deConfUSIon_apply_rgb2acr.m\`,	Automatic JM color/order preparation using \`rgb2acr.xlsx\` and \`list_selected_regions.txt\`.

Workflow	Main files	Notes
	`readFileList.m`, `atlas_tools/*`	
Segmentation	`Segmentation.m`	Atlas-based extraction of region time courses. Default baseline is 30-240 sec and entered in seconds.

Segmentation note: Segmentation in this package means atlas-label-based region extraction, not AI/image segmentation. The output includes MAT and CSV files such as `Segmentation_Both_zscore_*.csv`, `Segmentation_Both_raw_*.csv`, and `Segmentation_RegionTable_*.csv`.

13. Functional connectivity workflow

Functional connectivity is correlation-based and is sensitive to preprocessing, masks, global signal, motion, and atlas-label quality. Run FC only after QC/masking/registration decisions are stable.

7. Choose the active dataset version carefully. FC differs between raw, filtered, scrubbed, PCA/ICA-denoised, and motor-reconstructed data.
8. Load or create a valid brain/signal mask to exclude non-brain voxels and artifacts.
9. For atlas/ROI FC, load region labels and names that match the current functional space and slice.
10. For step-motor data, use the helper-supported workflows to locate per-slice region labels and names.
11. Use seed, ROI heatmap, pair, or graph modes as appropriate.
12. Inspect whether correlations are anatomically plausible or dominated by global/motion artifacts.
13. Export heatmaps, graphs, z maps, and source logs only after checking time courses and masks.

```
r = cov(x,y) / (std(x) x std(y))
Fisher_z = atanh(r)
```

FC statistics note: The current source includes JM ordering/color support through `deConfUSIon_fc_jm_order.m`, `deConfUSIon_prepare_atlas.m`, and atlas_tools files. Group FC code contains Fisher-z statistics patches, so correlations should be averaged/tested in z-space and converted back with tanh only for display.

14. Group analysis workflow

Group Analysis is the final summary module. It should be used after individual datasets are accepted and exported inputs are checked.

14. Prepare individual exports first: ROI time-course text files, SCM group bundles, segmentation tables, registration outputs, or FC bundles as appropriate.
15. Open Group Analysis only after checking active dataset, animal/session labels, acquisition mode, and preprocessing chain.
16. Add subjects/animals and verify group/condition labels manually.
17. For ROI analysis, load ROI exports and verify time axis, baseline/signal windows, and group assignment.
18. For map analysis, load exported SCM group bundles rather than raw Studio memory if possible.
19. For FC analysis, ensure correlation statistics are handled consistently and labels/regions match across animals.
20. Use outlier tools for inspection, not automatic deletion. Document every excluded animal/session.
21. Export Excel/CSV summaries, figures, PPT outputs, bundles, and the group-analysis session state.

Tip / Trick: Before group analysis, make a simple tracking table with every animal/session, acquisition mode, preprocessing steps, exclusions, final dataset label, and exported file paths. This prevents mixing raw, processed, and PSC versions.

15. Output files and naming

Folder	Typical contents	User action
QC/	QC PNGs and MAT summaries.	Review before accepting dataset; useful for supplements/debugging.
Preprocessing/	New MAT files from frame rejection, scrubbing, imregdemons, smoothing, filtering, PCA/ICA, despike, or motor.	Each step should create a new named dataset version.
PSC/	PSC MAT files and baseline metadata.	Inputs to SCM/video workflows.
Visualization/	SCM figures, ROI exports, videos, PPTs,	Main individual-level outputs.

Folder	Typical contents	User action
	underlays.	
Masks/	Brain/underlay masks, overlay/signal masks, mask bundles.	Keep masks tied to the dataset/slice they were created for.
Registration/	3D registration resources and transformations.	Dataset-specific, not source repository content.
Registration2D/	Reg2D files and 2D atlas underlays.	Used for 2D/step-motor atlas alignment.
Segmentation/	Segmentation MAT and CSV region-time outputs.	Use after registration/label-map validation.
FunctionalConnectivity/	FC maps, heatmaps, graphs, bundles, source logs.	Only after QC/masking and confound review.
GroupAnalysis/	Group Excel/CSV summaries, figures, PPT outputs, bundles.	Final cohort-level outputs.
Reports/	Manual summaries or generated reports.	Use for archive/review.

15.1 Dataset names

The current source contains several `deConfUSIon_*display*`, `*_canonical_dataset_label*`, and `*_safe_preproc_save_path*` helpers. Their purpose is to keep dropdown names readable while using shorter physical filenames when paths become too long. User-facing labels should preserve animal ID, scan ID, and the processing chain.

Good visible label example	Bad/ambiguous label example
1005_scan3_raw	dataset1
1005_scan3_raw_motor_pca_imregdemons	tmp_preproc_001
animal_scan_filter_band_0p01_0p20	filtered

16. Publication-ready QC checklist

QC item	Acceptable result	Action if failed
Raw movie viewed	No severe bubble/shadow/motion artifact; vascular structure visible.	Reacquire or exclude if severe.
TR confirmed	Total time matches acquisition log and stimulus/injection timing.	Reload and correct TR before analysis.
Baseline stable	No motion, injection artifact, or drift during baseline.	Move baseline or reject dataset if no clean baseline exists.
Frame outliers reviewed	Outlier count small and interpolation justified.	Document rejection/scrubbing settings.
Preprocessing justified	Every processing step has a QC reason.	Avoid inconsistent preprocessing across animals.
Filtering validated	Filtered output preserves anatomy/mean image and QC plots look reasonable.	Check restoreMean, cutoff, Nyquist, and static image handling.
SCM inspected	Map anatomically and temporally plausible, not edge/bubble/global artifact.	Verify with video and ROI trace.
Mask/underlay verified	Mask aligns with displayed slice and underlay.	Reload/regenerate mask/underlay if mismatch.
Registration inspected	Atlas/underlay match visually.	Do not trust atlas segmentation if registration is poor.
Segmentation checked	Region labels/time courses match expected anatomy/slices.	Fix label map, registration, or slice matching before export.
FC confounds reviewed	Correlations not dominated by motion, global signal, or non-brain voxels.	Use masks/QC/confound strategy before interpretation.
Group labels checked	Animal/session/condition labels correct.	Fix before statistics or export.
Studio Log exported	Analysis decisions saved.	Export before closing session.

17. Troubleshooting

Problem	Likely cause	Fix
Toolbox does not launch	Missing split GUI files or path problem.	Run from toolbox root. Use `run_fusi_studio` or `deConfUSIon`, not old `fusi_studio`. Confirm `fusi_studio_GUI.m` and `fusi_studio_callback.m` exist.
Dataset does not load	Unsupported variable name, corrupt MAT/NIfTI, huge file, missing reader, wrong path.	Use `whos -file` or `load` to inspect variables. Ensure a valid 3D/4D image stack exists.
Wrong total time	Incorrect TR or suspicious metadata.	Reload and confirm TR from acquisition log. Loader stores detection source and imputed/default status.
Buttons remain disabled	Data not loaded or GUI state not updated.	Load fUSI data again and inspect Studio Log for errors.
SCM map looks wrong	Bad baseline, wrong active dataset, motion, wrong underlay, mask mismatch.	Check dropdown, baseline, raw movie, ROI trace, and mask/underlay slice.
Filtered image looks dark/low-res	Older filtering behavior or mean not restored.	Use current `filtering.m` from this ZIP. It restores voxelwise mean by default; confirm `restoreMean=true` and inspect filtering QC PNGs.

Problem	Likely cause	Fix
Filtering errors for 2D/static data	Static 2D image or unsupported dimensionality.	Current `filtering.m` returns static 2D unchanged and supports [Y X T] and [Y X Z T]. Verify you are not accidentally calling an old backup version.
imregdemons fails	Image too small, invalid values, too aggressive settings, wrong mode for step-motor.	Use fewer/smaller settings, median nsub preset, or slice-wise step-motor mode; skip if data are stable.
Motor reconstruction wrong	Wrong frames per slice/position or inconsistent split filenames.	Verify frame counts and slice_t naming before reconstruction.
PCA/ICA removes response	Biological response selected as artifact component.	Return to pre-PCA/ICA dataset and rerun with fewer/no components.
Segmentation wrong slice/labels	Reg2D/label map not matched to functional slice.	Check source slice, atlas slice, label map dimensions, and region-name file.
FC globally high	Global signal, motion, or non-brain voxels dominate.	Use masks, QC, and a confound strategy before interpreting FC.
Old behavior appears	Backup/archived file earlier on MATLAB path.	Remove backups from active path and run `which functionName -all`.

18. Source-package cleanup and release notes

The uploaded ZIP contains a much cleaner active root than older packages, but also includes `.git`, `backups`, archived patch scripts, compressed cleanup packages, and the old manual PDF. These are useful for history/recovery but should be separated from a normal user release.

Topic	Recommendation
Keep active runtime path clean	Do not add `backups/` recursively. Only root active files and `atlas_tools/` should be on the normal MATLAB path.
Backups folder	Current final cleanup notes say to keep backups until full testing with normal 2D and 2D step-motor data passes. After validation, copy/zip backups outside the repository and remove from release ZIP.
atlas_tools	Keep. Contains `rgb2acr.xlsx`, `list_selected_regions.txt`, and manual helpers needed for JM color/order support.
run_fusi_studio.m	Keep. Required to assemble the split GUI/runtime.
deConfUSIon.m	Keep. User-facing convenience launcher.
studio_load_options_dark_dialog.m	Keep. Current runtime load-options dialog; replaces old patch16 name.
Dropdown metadata helpers	Keep. Needed for saved preprocessing outputs and full display names.
Popup/autofit helpers	Keep. GUI/timer callback helpers remain external by design.
FC/step-motor helpers	Keep. Shared across FunctionalConnectivity, Studio, segmentation, and step-motor workflows.
Generated outputs	Do not commit QC/Preprocessing/PSC/Visualization/Registration/GroupAnalysis outputs except small curated examples.
Release package	Ship a clean ZIP with active code, atlas resources or download instructions, README, example data, and this updated manual. Exclude `.git`, backups, generated outputs, and old patch archives.

Appendix A. Current MATLAB file inventory

This inventory is derived from the uploaded ZIP after excluding `.git`, backups, compressed backup packages, and generated files. It lists 82 active MATLAB files: 80 in the root and 2 in `atlas_tools/`.

Category	Number of files
Acquisition / trigger controller	3
Atlas / registration / segmentation	14
Core launch / GUI / loader	12
Functional connectivity	8
Group analysis	5
Preprocessing	8
Quality control	3
Visualization / PSC / masks	9
deConfUSion compatibility / cleanup helper	20

File	Category	Lines	Purpose from header/signature
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_COMMAND.m	Acquisition / trigger controller	3371	ASCII safe, MATLAB 2017b compatible Backend runner used by the GUI entry file. WHAT THIS VERSION DOES
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_OBJECT.m	Acquisition / trigger controller	775	ASCII safe MATLAB 2017b compatible Per-frame callback object for:
vfUSI_StimBox_TTL_EACH_FRAME_OR_TRIGGER_ACCESSORIES_PARAMETERS.m	Acquisition / trigger controller	2692	UTF-8 safe, MATLAB 2017b compatible GUI ENTRY SCRIPT Keep this filename unchanged because the encrypted launcher may call it.
atlas_tools/ deConfUSIon_reorder_FC_by_list.m	Atlas / registration / segmentation	21	deConfUSIon_reorder_FC_by_list JM example wrapper for FC matrices.
atlas_tools/save_correct_colors.m	Atlas / registration / segmentation	37	save_correct_colors Add/fix atlas.infoRegions.rgb from JM rgb2acr.xlsx. Usage: atlas = save_correct_colors('allen_brain_atlas.mat','rgb2acr.xlsx');
coreg.m	Atlas / registration / segmentation	852	coreg.m Unified atlas-registration launcher Updated robust version
coreg_3d.m	Atlas / registration / segmentation	795	=== fUSI Studio - Atlas Coregistration ASCII only
coreg_coronal_2d.m	Atlas / registration / segmentation	1455	coreg_coronal_2d.m Simple manual 2D coronal atlas registration launcher. Updated for 2D step-motor / multi-slice data:
deConfUSIon_find_stepmotor_seg_fc_files.m	Atlas / registration / segmentation	206	Auto-detect step-motor Reg2D, Segmentation, label-map and region-name files. MATLAB 2017b compatible.
deConfUSIon_prepare_atlas.m	Atlas / registration / segmentation	57	deConfUSIon_prepare_atlas Automatic JM atlas color/order preparation. This does not change atlas geometry. It adds corrected rgb fields and stores selected/reordered region indices in atlas.deConfUSIon.
interpolate3D.m	Atlas / registration / segmentation	90	interpolate3D (ROBUST) -- Paper-faithful intent:
mapscan.m	Atlas / registration / segmentation	110	Urban Lab - NERF empowered by imec, KU Leuven and VIB Mace Lab - Max Planck institute of Neurobiology Authors: G. MONTALDO, E. MACE
moveimage.m	Atlas / registration / segmentation	286	=== Robust image mover for HUMoR registration GUI MATLAB 2017b compatible, ASCII-safe
readFileList.m	Atlas / registration / segmentation	134	readFileList Parse JM selected-region list and map acronyms to atlas indices. Usage: D = readFileList('list_selected_regions.txt', atlas.infoRegions);
registration_ccf.m	Atlas / registration / segmentation	2016	=== Atlas registration GUI ASCII only
registration_coronal_2d.m	Atlas / registration / segmentation	1784	registration_coronal_2d.m Manual simple 2D coronal atlas registration. Updated for 2D step-motor / multi-slice data:
Segmentation.m	Atlas / registration / segmentation	2188	Segmentation.m HUMoR fUSI Studio atlas-based segmentation / region-time extraction. MATLAB 2017b + 2023b compatible, ASCII-safe.
addLog.m	Core launch / GUI / loader	86	addLog.m - fallback logger for deConfUSIon / fUSI Studio MATLAB 2017b compatible.
buildFooterLabel.m	Core launch / GUI / loader	8	s = buildFooterLabel()
datasetDropdownCallback.m	Core launch / GUI / loader	34	datasetDropdownCallback(src,~)

File	Category	Lines	Purpose from header/signature
deConfUSIon.m	Core launch / GUI / loader	17	deConfUSIon - main launcher for deConfUSIon / fUSI Studio. This directly calls run_fusi_studio because the split GUI runtime still depends on fusi_studio_GUI.m + fusi_studio_callback.m assembly.
fusi_studio_callback.m	Core launch / GUI / loader	5874	fusi_studio_callback - callback/helper source part 2 of the split Studio This is a valid MATLAB file that stores one source chunk for the split deConfUSIon / fUSI Studio. Run run_fusi_studio.m to assemble and launch.
fusi_studio_GUI.m	Core launch / GUI / loader	6205	fusi_studio_GUI - GUI/source part 1 of the split Studio This is a valid MATLAB file that stores one source chunk for the split deConfUSIon / fUSI Studio. Run run_fusi_studio.m to assemble and launch.
loadDataCallback.m	Core launch / GUI / loader	295	loadDataCallback(src,~)
loadFUSIData.m	Core launch / GUI / loader	921	loadFUSIData --- ----- Robust fUSI data loader with TR / time inference
run_fusi_studio.m	Core launch / GUI / loader	155	run_fusi_studio - clean launcher for split deConfUSIon / fUSI Studio. MATLAB 2017b + 2023b compatible. Source files kept in toolbox root:
studio_load_options_dark_dialog.m	Core launch / GUI / loader	571	studio_load_options_dark_dialog Dark modern Load Dataset popup for HUMoR / fUSI Studio. MATLAB 2017b + 2023b compatible.
studio_mkdir.m	Core launch / GUI / loader	6	studio_mkdir(p)
studio_resolve_paths.m	Core launch / GUI / loader	175	P = studio_resolve_paths(par, fileLabel, exportRootOverride)
deConfUSIon_FC_find_stepmotor_txt_names.m	Functional connectivity	237	deConfUSIon_FC_find_stepmotor_txt_names Recursively finds readable atlas/region-name TXT/CSV/TSV/MAT files for step-motor FC.
deConfUSIon_FC_force_layout.m	Functional connectivity	309	Stable/idempotent FC GUI layout fixer. Repeated calls produce the same layout. No timers, no callbacks, no wrappers.
deConfUSIon_fc_jm_order.m	Functional connectivity	85	deConfUSIon_fc_jm_order Reorder FC ROI names according to JM list. The function returns ord into the input arrays. Unmatched regions are kept after matched regions in their original order.
deConfUSIon_FC_make_slice_roi_result.m	Functional connectivity	234	True slice-specific ROI FC result. Recomputes ROI time courses from selected Z slice and preserves L/R labels.
deConfUSIon_FC_read_region_names_file.m	Functional connectivity	241	Robust FC region-name reader. Supports files like AtlasRegions_slice111.txt, CSV/TSV, MAT, Segmentation_*.mat.
deConfUSIon_FC_remember_layout.m	Functional connectivity	129	Capture/restore the good FC GUI layout so Load Seg MAT does not shrink fonts.
deConfUSIon_FC_stepmotor_read_folder.m	Functional connectivity	263	Recursively finds step-motor region TXT/CSV/MAT names and per-slice atlas labels.
FunctionalConnectivity.m	Functional connectivity	8449	FunctionalConnectivity.m fUSI Studio - Functional Connectivity GUI MATLAB 2017b compatible, ASCII-only.
GroupAnalysis.m	Group analysis	7536	GA_FISHERZ_STATS_PATCH_20260512 FC group matrices are averaged/statistically compared in Fisher z space. Convert back with tanh(Z) only for Pearson-r display if needed.
GroupAnalysis_Common.m	Group analysis	5021	Print full module errors in Command Window.
GroupAnalysis_FC.m	Group analysis	4989	GA_FISHERZ_STATS_PATCH_20260512 FC group matrices are averaged/statistically compared in Fisher z space. Convert back with tanh(Z) only for Pearson-r display if needed.
GroupAnalysis_Map.m	Group analysis	1218	GA_AUTO_ERROR_PRINT_PATCH_V1 GroupAnalysis_Map - self-contained map backend for modular GroupAnalysis. MATLAB 2017b + 2023b compatible.
tryReadSCMroiExportTxt.m	Group analysis	48	Tries to read SCM_gui ROI export txt: - header lines begin with '#' - table columns: time_sec time_min PSC
despike.m	Preprocessing	132	=== VOXEL-WISE MAD DESPIKING (STABLE VERSION) ===
filtering.m	Preprocessing	563	=== fUSI Studio - Robust Butterworth Filtering Engine ==

File	Category	Lines	Purpose from header/signature
			===========
ica_denoise.m	Preprocessing	1512	ICA_DENOISE V12 — integrated all-slice / slice-specific recompute GUI
imregdemons_preprocess.m	Preprocessing	510	imregdemons_preprocess Supports: - 3D input: Iin [Y X T]
motor.m	Preprocessing	1709	============================== ========================= MOTOR RECONSTRUCTION (FRAME-BASED, TWO MODES, STUDIO-SAFE) Supports:
pca_denoise.m	Preprocessing	1050	PCA_DENOISE V12 — integrated all-slice / slice-specific recompute GUI The first PCA/ICA popup only chooses method. This GUI handles slice scope.
scrubbing.m	Preprocessing	589	============================== ========================== SCRUBBING (MEMORY-SAFE) - Detection: DVARS or Global Signal
temporalsmoothing.m	Preprocessing	270	temporalsmoothing Two modes in one file: 1) Sliding temporal smoothing
frameRateQC.m	Quality control	166	frameRateQC Single-window diagnostic frame-rate / rejected-volume QC. Shows ONE combined QC figure per run:
interpolateRejectedVolumes.m	Quality control	42	interpolateRejectedVolumes -- ----- Interpolate rejected volumes along time (Urban/Montaldo style)
qc_fusi.m	Quality control	2063	qc_fusi -- fUSI Studio QC engine (MATLAB 2017b) Keeps existing QC plots and ADDS paper-style / Python-style QC: - Burst error QC (ratio map + noisy voxels + coverage over time)
blackbdy_iso.m	Visualization / PSC / masks	287	blackbdy_iso FSLeyes "blackbdy_iso" colormap (0..1 RGB) Usage: cm = blackbdy_iso(); % 256
computePSC.m	Visualization / PSC / masks	269	computePSC (UPDATED: supports 2D + matrix probe 3D) -- ----- Classic PSC pipeline.
fUSI_Live_Studio.m	Visualization / PSC / masks	2439	fUSI_Live_Studio Clean GUI refresh of the live viewer. Same overall functionality, improved styling/layout only.
mask.m	Visualization / PSC / masks	3543	mask.m - fUSI Studio Mask Editor MATLAB 2017b / 2023b compatible PURPOSE
play_fusi_video_final.m	Visualization / PSC / masks	6361	============================== ========================= fUSI Video GUI (MATLAB 2023b) GUI-cleaned version
SCM_gui.m	Visualization / PSC / masks	6625	SCM_gui - fUSI Studio SCM viewer/controller MATLAB 2017b + 2023b compatible, ASCII-safe. Expected PSC dimensions:
shortenMiddle.m	Visualization / PSC / masks	55	shortenMiddle Safely shorten long labels/paths by preserving beginning and end. Needed by SCM / Video setup dialogs and dropdown labels.
showScmVideoSetupDialog.m	Visualization / PSC / masks	1024	UNDERLAY CHOICES 1 = Default (current SCM / Video bg) 2 = Mean of ACTIVE dataset
winter_brain_fsl.m	Visualization / PSC / masks	286	winter_brain_fsl FSLeyes-derived winter brain colormap. Returns an n x 3 RGB colormap in [0,1].
deConfUSIon_add_preproc_lazy_datasets.m	deConfUSion compatibility / cleanup helper	283	Scanner for saved Preprocessing/PMATs. Repairs abbreviated names.
deConfUSIon_apply_rgb2acr.m	deConfUSion compatibility / cleanup helper	145	deConfUSIon_apply_rgb2acr Apply RGB colors from JM rgb2acr.xlsx to atlas.infoRegions.
deConfUSIon_best_visible_dataset_name.m	deConfUSion compatibility / cleanup helper	47	Robust visible dataset name for Studio dropdowns.
deConfUSIon_canonical_dataset_label.m	deConfUSion compatibility / cleanup helper	400	V25 compatibility wrapper: preserve ordered processing chain.
deConfUSIon_commit_full_display_name.m	deConfUSion compatibility / cleanup helper	19	Append exact full display name into top-level MAT metadata.
deConfUSIon_compact_chain_name.m	deConfUSion compatibility / cleanup helper	40	Clean user-facing dataset chain names without deleting scan IDs.
deConfUSIon_display_from_file_context.m	deConfUSion compatibility / cleanup helper	51	Build readable name for saved/lazy preprocessing files when metadata is missing.
deConfUSIon_display_name_from_sources.m	deConfUSion compatibility / cleanup helper	362	Build readable Studio dropdown labels without losing animal/scan identity. Examples: 1005_scan3_raw
deConfUSIon_fix_processing_name.m	deConfUSion compatibility / cleanup helper	301	Ensure user-facing dataset names preserve applied processing steps. Especially fixes: motor -> PCA -> imregdemons where PCA was missing after reload.
deConfUSIon_fix_scm_video_dialog_fonts.m	deConfUSion compatibility / cleanup helper	158	Final SCM/Video setup popup layout helper.
deConfUSIon_fix_studio_dataset_names.m	deConfUSion compatibility / cleanup helper	44	Repair Studio dataset display metadata and preserve new preprocessing outputs

File	Category	Lines	Purpose from header/signature
			in dropdown.
deConfUSIon_force_fullscreen_fig.m	deConfUSion compatibility / cleanup helper	47	deConfUSIon_force_fullscreen_fig Opens normal MATLAB figure GUIs in a large/maximized window. MATLAB 2017b + 2023b compatible.
deConfUSIon_is_bad_display_name.m	deConfUSion compatibility / cleanup helper	14	True if visible name looks like an internal temporary/preproc key.
deConfUSIon_make_loaded_display_name.m	deConfUSion compatibility / cleanup helper	26	Visible RAW label for Studio dropdown. Preferred: animal_scanX_raw, e.g. 1005_scan3_raw.
deConfUSIon_pcaica_scope_tag.m	deConfUSion compatibility / cleanup helper	11	tag = deConfUSIon_pcaica_scope_tag(scopeInfo)
deConfUSIon_popup_autofit_apply.m	deConfUSion compatibility / cleanup helper	11	Direct wrapper for HUMoR popup auto-fit.
deConfUSIon_popup_autofit_timer.m	deConfUSion compatibility / cleanup helper	90	deConfUSIon_popup_autofit_timer Stable version: auto-polishes general setup popups, but NEVER touches Atlas Segmentation windows. This prevents segmentation popup vibration.
deConfUSIon_popup_polish_now.m	deConfUSion compatibility / cleanup helper	393	deConfUSIon_popup_polish_now NO-VIBRATE PATCH: do not call HUMoR_segmentation_popup_fix_now() here. That second helper resizes the segmentation popup to a different size,
deConfUSIon_safe_preproc_save_path.m	deConfUSion compatibility / cleanup helper	120	Short physical filename to avoid Windows/network path errors. The readable chain is stored in displayNameFull/preprocDisplayName.
deConfUSIon_write_full_display_metadata.m	deConfUSion compatibility / cleanup helper	30	Append full display metadata into a saved MAT file.

Appendix B. Change log for this manual update

Date	Change
2026-06-09	Regenerated manual from the newly uploaded `HUMOR-Analysis-Tool (2) (9).zip` package.
2026-06-09	Renamed toolbox throughout the manual from HUMoR/HUMOR to deConfUSion while preserving legacy-name notes for old paths, README text, and archived files.
2026-06-09	Updated launch instructions: active user launchers are `run_fusi_studio.m` and `deConfUSIon.m`; root-level `fusi_studio.m` is not present in the current ZIP.
2026-06-09	Updated architecture description: `run_fusi_studio.m` assembles `fusi_studio_GUI.m` + `fusi_studio_callback.m` into a temporary `fusi_studio_runtime.m`.
2026-06-09	Updated GUI section table to the current 10-section layout, including placeholder status for GLM/Regression, Velocity Analysis, and Community Analysis.
2026-06-09	Retained and expanded the abbreviations/key-terms section.
2026-06-09	Updated data loading notes: TR detection source tracking, `TRWasImputed` handling, 2D default TR 0.320 s, matrix/3D default TR 0.480 s.
2026-06-09	Updated filtering documentation for current robust patch: supports [Y X T] and [Y X Z T], returns static 2D unchanged, restores voxelwise mean, writes higher-resolution QC PNGs.
2026-06-09	Updated imregdemons documentation for median/mean block mode, nsub/regSmooth setup, true 3D versus slice-wise step-motor support.
2026-06-09	Updated Motor documentation for current frame-based continuous and split reconstruction logic.
2026-06-09	Updated segmentation documentation: current `Segmentation.m` is active atlas-based region-time extraction with CSV/MAT outputs.
2026-06-09	Updated FC documentation for step-motor label/name helpers, layout stabilization helpers, JM atlas order/color support, and Fisher-z group statistics notes.
2026-06-09	Updated source cleanup guidance using current final cleanup decision logs: keep `atlas_tools`, `run_fusi_studio`, `studio_load_options_dark_dialog`, dropdown metadata helpers, popup/autofit helpers, and FC/step-motor helpers; keep backups only until full testing passes.
2026-06-09	Rebuilt Appendix A inventory from the current source package: 80 root MATLAB files and 2 atlas_tools MATLAB files, excluding backups and `.git`.

Source audit notes

Audit item	Finding
Files inspected	82 active MATLAB files after excluding backups and .git.
Old root GUI file	`fusi_studio.m` is referenced in older README text but is not present at the current root. The split runtime is the active architecture.
Old load-options patch name	`studio_load_options_dark_dialog_patch16.m` is referenced in older README text but current root has `studio_load_options_dark_dialog.m`.
Old helper references	Several old helper names in the README are not present at root in this ZIP. This manual uses the current active files instead.
Large atlas resource	`allen_brain_atlas.mat` is in the uploaded ZIP but was excluded from text inspection due to size. Keep it available for atlas workflows.